

Arakawa–Schubert (1974): goal of the paper

1

Large-scale models predict temperature and humidity averaged over large areas. They do not resolve the cumulus clouds inside, yet the clouds heat, cool, dry and moisten the air.

Goal

Compute the collective effect of the clouds from the large-scale fields themselves, so that the large-scale equations close. This is *cumulus parameterization*.

Before this paper

Ooyama (1971) reduced the problem to one cloud-base function, but left that function undetermined. This paper determines it.

How: three steps

Step 1 · §2–3 · slide 2

Budgets of the air between the clouds: the clouds act only through three profiles in z .

Step 2 · §4–5 · slide 3

Clouds as entraining plumes of types λ : the three profiles follow from one spectrum.

Step 3 · §6–7 · slides 4–5

Quasi-equilibrium: the large-scale build-up of buoyancy balances its use by the clouds.

Unknowns left

3 profiles
 $M_c, D, \hat{l}$

1 spectrum
 $\mathcal{M}_B(\lambda)$

none:
closed

In a model, each time step: $\bar{s}, \bar{q} \rightarrow \mathcal{M}_B(\lambda) \rightarrow M_c, D, \hat{l} \rightarrow \bar{s}, \bar{q}$ at the next step

Paper review: Arakawa & Schubert (1974) | Slide 1

Arakawa–Schubert (1974): goal of the paper

Interaction of a Cumulus Cloud Ensemble with the Large-Scale Environment, Part I, Arakawa and Schubert (1974), J. Atmos. Sci. 31, 674–701. A large-scale disturbance contains many cumulus clouds whose time and space scales are much smaller than its own. Because of this scale separation, the paper aims to predict the large-scale fields by describing only the collective influence of the clouds, not each cloud. The paper calls this the goal of cumulus parameterization. The result is meant for prognostic models of the large-scale atmosphere, such as the UCLA general circulation model.

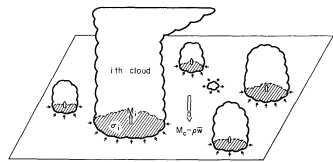
The reduction has three steps. First, budgets for the environment show that the clouds change its dry static energy $\bar{s}$ and water vapour $\bar{q}$ only through three profiles: the total cloud mass flux $M_c(z)$, the total detrainment $D(z)$ and the liquid water $\hat{l}(z)$ in the detraining air. In the subcloud mixed layer they act only on its depth. Second, a spectral ensemble of entraining plumes computes all three profiles from the environment and a single unknown function, the mass flux $\mathcal{M}_B(\lambda)$ of each cloud type at the top of the mixed layer. Third, the quasi-equilibrium of the cloud work function gives an integral equation for $\mathcal{M}_B(\lambda)$, and the system is closed. Slide 5 shows the observational test of quasi-equilibrium.

The theory describes a two-way interaction: the clouds modify the environment (step 1), and the environment controls the clouds (steps 2 and 3). Earlier parameterizations were largely empirical. Ooyama (1971) had reduced the problem to a cloud-base dispatcher function but left its determination open, so his parameterization was not closed. The bottom line shows how the closed theory is used in a model. Source: Abstract, Sections 1 and 8.

Step 1: clouds enter through three profiles

2

Budgets of the air between clouds: the model knows every term except M_c , D and $\hat{l}$.



$$\begin{aligned}\rho \partial_t \bar{s} &= D(\hat{s} - \bar{s} - L\hat{l}) + M_c \partial_z \bar{s} + LS_s \\ \rho \partial_t \bar{q} &= D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} + LS_q\end{aligned}$$

detrainment: subsidence: large scale
cools, moistens warms, dries

Mixed layer below: clouds only lower its top z_B , through $M_B = M_c(z_B)$.

Symbols

$s = c_p T + gz$ dry static energy, J kg^{-1}

q water-vapour mixing ratio, $\text{kg vapour} / \text{kg dry air}$

$\bar{s}, \bar{q}$ large-scale means, i.e. the environment

$M_c = \sum_i \rho \sigma_i w_i$ cloud mass flux, $\text{kg m}^{-2} \text{s}^{-1}$

σ_i, w_i area fraction, vertical velocity of cloud i

D mass detrained per unit volume and time

$\hat{s}, \hat{q}^*$ in the detrained air, saturated; set by $\bar{s}, \bar{q}, \hat{l}$

$\hat{l}$ liquid water in the detrained air

L, ρ latent heat of vaporization; air density

LS large-scale advection (+ radiation for s)

Unknowns left: three profiles, $M_c(z)$, $D(z)$ and $\hat{l}(z)$

Step 1: clouds enter through three profiles

These are the equations the large-scale model must integrate. All terms are known to it except the cloud terms. Cloud i covers the area fraction σ_i and carries the mass flux $\rho \sigma_i w_i$. The sum M_c drives compensating environmental subsidence. In a stably stratified atmosphere ($\partial_z \bar{s} > 0$) this warms, and since $\partial_z \bar{q} < 0$ it dries. Detrainment supplies cloud air and liquid water, whose evaporation cools ($-L\hat{l}$) and moistens. The latent heat released inside the clouds does not warm the environment directly: it keeps the clouds buoyant and so maintains M_c and the subsidence warming.

The variables are the dry static energy $s = c_p T + gz$, conserved by a dry parcel moving adiabatically, and the water-vapour mixing ratio q . Hats denote detraining-air values, the star denotes saturation, and L is latent heat. $\hat{s}$ and $\hat{q}^*$ are not further unknowns: Eqs. (72)–(73) express them through $\bar{s}, \bar{q}, \bar{q}^*$ and $\hat{l}$. For legibility the slide groups the large-scale terms as $LS_s = -\rho \bar{\mathbf{v}} \cdot \nabla \bar{s} - \rho \bar{w} \partial_z \bar{s} + \bar{Q}_R$ and $LS_q = -\rho \bar{\mathbf{v}} \cdot \nabla \bar{q} - \rho \bar{w} \partial_z \bar{q}$. $\bar{Q}_R$ is the radiative heating, including that inside the clouds, which the paper sets aside.

In the subcloud mixed layer the clouds do not act on temperature and moisture directly. They drain mass through its top, and this cumulus-induced subsidence makes it shallower (Eq. 126). The equations assume a small active cloud fraction and a horizontally uniform environment. Figure 1 is the paper's conceptual sketch, not a measured cloud map. Section 4 opens with the conclusion on the slide: the parameterization problem is reduced to finding $M_c(z)$, $D(z)$ and $\hat{l}(z)$. Source: Sections 2–4, Figure 1, Equations (7), (12), (72)–(75) and (126).

Step 2: one spectrum gives all three profiles

3

Each cloud type λ mixes in outside air at every height, at its own rate. Given the environment, each type follows, and summing the types gives M_c , D and $\hat{l}$.

One cloud type λ

$$\partial_z \eta = \lambda \eta \Rightarrow \eta = e^{\lambda(z-z_B)}$$

$$\eta h_c = h_M + \lambda \int_{z_B}^z \eta \bar{h} dz'$$

All types together

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

$$D(z) = -\mathcal{M}_B(\lambda_D) \eta(z, \lambda_D) d\lambda_D / dz$$

$$\hat{l}(z) = l(z, \lambda_D)$$

Symbols

λ fractional entrainment rate, km^{-1} : the type label. Large λ : small, diluted cloud with a low top

$\eta(z, \lambda)$ mass flux of type λ , normalized to 1 at z_B

$h = s + Lq$ moist static energy: conserved also when vapour condenses

h_c, h_M h inside the cloud; in the subcloud mixed layer

z_B mixed-layer top: clouds start here with mixed-layer air

$z_D(\lambda)$ cloud top, where buoyancy vanishes: $h_c = \bar{h}^*$, the saturated h of the environment

$\lambda_D(z)$ type whose top is at height z

$l(z, \lambda)$ liquid water in type λ ; needs a rain model

$\mathcal{M}_B(\lambda)$ mass flux of type λ at z_B

Unknown left: one function, the spectrum $\mathcal{M}_B(\lambda)$

Step 2: one spectrum gives all three profiles

The ensemble is divided into entraining plumes with fractional entrainment rate λ , assumed constant with height. Each plume starts at z_B with mixed-layer air and mixes in environmental air at every height up to its top, where all its mass detrains. The normalized mass flux satisfies $\partial_z \eta = \lambda \eta$ and $\eta(z_B) = 1$, so it grows exponentially up to the plume top. The reference level z_B is the mixed-layer top, which can lie below cloud base. Through the tower model of Simpson, λ is inversely proportional to the cloud radius: large λ means small clouds.

The moist static energy $h = s + Lq$ follows the entraining-plume budget. On the slide, the arguments are suppressed: $\eta(z, \lambda) h_c(z, \lambda) = h_M + \lambda \int_{z_B}^z \eta(z', \lambda) \bar{h}(z') dz'$. Entrainment dilutes mixed-layer air with environmental air, which has lower h in the mid-troposphere. The cloud top z_D is the level of vanishing buoyancy, $h_c(z_D) = \bar{h}^*(z_D)$, approximately $\bar{h}^*(z_D)$. Figure 8 shows the resulting cloud-top pressure for a tropical sounding: increasing entrainment gives a lower top.

Summing the types gives the three profiles of step 1. $M_c(z)$ sums the types still rising at z , those with $\lambda < \lambda_D(z)$ (Eq. 77). $D(z)$ is the mass flux of the types whose top lies at z (Eq. 79). $\hat{l}(z)$ is the liquid water of the type whose top is at z (Eq. 91). Computing $l(z, \lambda)$ requires a rainfall parameterization, which the paper calls very crude. Cloud-base values come from the mixed layer: $h_c(z_B) = h_M$ (Eqs. 129–131). Everything is then fixed by the environment except the mass-flux density $\mathcal{M}_B(\lambda)$, which is not a number or radius density. Source: Sections 4–5, Equations (77)–(97) and (129)–(131), Figure 8.

Step 3: quasi-equilibrium fixes the spectrum

4

No theory gives the number, lifetime and mass flux of the clouds of each type. The paper balances energy instead: the large scale builds up $A(\lambda)$, the clouds use it up.



$$A(\lambda) = \int_{z_B}^{z_D(\lambda)} \eta \underbrace{\frac{g(s_{vc} - \bar{s}_v)}{c_p \bar{T}}}_{\text{buoyancy}} dz$$

$$\frac{dA}{dt} = \int K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) \simeq 0$$

$\simeq 0$: quasi-equilibrium, clouds adjust much faster than F changes.

Symbols

$A(\lambda)$ cloud work function: buoyant kinetic energy of type λ per unit mass flux

$s_v = s + c_p \bar{T}(0.608q - l)$, virtual static energy: vapour lightens the air, liquid water l weighs it

s_{vc} , $\bar{s}_v$ in the cloud; in the environment

g , $\bar{T}$ gravity; environment temperature

$A > 0$ type λ can grow; $A(0) \approx \text{CAPE}$

$K(\lambda, \lambda')$ change of $A(\lambda)$ per unit $\mathcal{M}_B(\lambda')$; mostly negative

$F(\lambda)$ change of A by large-scale processes

No unknowns left: $\mathcal{M}_B(\lambda)$ solves an integral equation, and the system is closed

Step 3: quasi-equilibrium fixes the spectrum

Section 7 first explains why a closure is needed. The spectrum is $\mathcal{M}_B(\lambda) = \mathcal{N}(\lambda) m_B(\lambda) / \tau(\lambda)$ (Eq. 145): the number of clouds of type λ , divided by their lifetime, times the mass a single cloud carries through z_B over its whole life. In 1974 neither theory nor data determined these three separately, and one-dimensional cloud models need cloud-base conditions as an input. Quasi-equilibrium makes the separation unnecessary.

The cloud work function is the buoyancy integral weighted by normalized mass flux. It measures kinetic-energy generation per unit mass flux at the reference level: the kinetic-energy budget is $dK/dt = A\mathcal{M}_B - \mathcal{D}$. The environment determines A , including the mixed-layer contribution. Positive A permits buoyant energy production but does not by itself prescribe cloud counts or triggering. For non-entraining clouds, $\lambda = 0$, the integral is close to what is now called CAPE; entraining types also need a moist environment.

The map shows the feedback: large-scale forcing changes A , buoyancy supports cloud activity, and the ensemble modifies the environment and thus its own work function. Net stabilization dominates the adjustment. Individual entries of K need not all be negative, since detrainment by shallow clouds can help deeper clouds.

The tendency equation separates the cloud contribution, $\int_0^{\lambda_{\max}} K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda'$, from the large-scale forcing F : cooling by large-scale ascent, advection, radiation and surface fluxes. Under quasi-equilibrium the tendency is small compared with these two contributions, giving a Fredholm integral equation of the first kind for $\mathcal{M}_B(\lambda)$. The equality holds for active cloud types, with a nonnegative spectrum and an inequality for absent types. This closes the system and is the principal result of Part I. It does not impose a steady temperature or moisture profile. Source: Sections 6–7, Equations (132)–(133), (142), (145), (150) and (158).

Step 3 tested: observations support quasi-equilibrium

5

Test with tropical data. If clouds did not respond, A would change as fast as the forcing (dashed line). The observed change of A stays near zero.

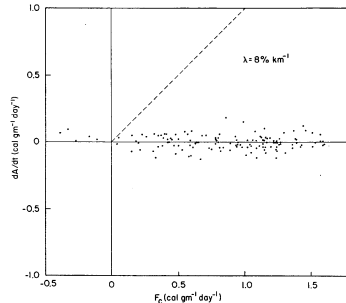


Fig. 13: observed dA/dt against F_C , for $\lambda = 8\% \text{ km}^{-1}$.

Two timescales

$$\tau_{\text{adj}} \sim 10^3\text{--}10^4 \text{ s}$$

15 min to 3 h

$$\tau_{\text{LS}} \gtrsim 10^5 \text{ s}$$

a day or more

Observed:

$$|dA/dt| \ll |F_C|$$

Symbols

F_C change of A that large-scale processes in the cloud layer would produce alone

dA/dt observed change of A

τ_{adj} time for the clouds to remove A

τ_{LS} time over which the large-scale forcing changes

cal g^{-1} energy per unit mass; 1

$\text{cal g}^{-1} \approx 4.2 \text{ kJ kg}^{-1}$

Clouds nearly cancel the forcing: A changes much less than F_C predicts

Step 3 tested: observations support quasi-equilibrium

The paper estimates a cloud adjustment time of $10^3\text{--}10^4 \text{ s}$ and a large-scale forcing time of at least 10^5 s . This separation motivates dropping the relatively small work-function tendency in the closure equation. The estimates are orders of magnitude, not measured lifetimes of individual thermals.

Figure 13 compares the observed tendency of the cloud work function with cloud-layer forcing F_C from the Marshall Islands data of Yanai et al. (1973). The upper panel, reproduced here, uses $\lambda = 8\% \text{ km}^{-1}$. Without cloud adjustment, the observations would lie along the dashed line $dA/dt = F_C$. Instead, the tendency is generally much smaller than the forcing, which reaches about $1.5 \text{ cal g}^{-1} \text{ day}^{-1}$. The lower panel for $\lambda = 16\% \text{ km}^{-1}$ gives the same qualitative result.

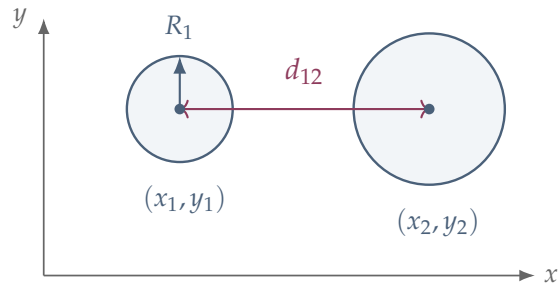
The test supports quasi-equilibrium in this setting. It omits the mixed-layer forcing F_M , which could not be evaluated from the conventional observations. Figure 14 separately shows that temperature and moisture profiles evolve while their combined work function remains small. Numerical solution methods and applications of the closure are deferred to Part II. Source: Section 7, Equations (146)–(154), Figures 13–14, and Section 8.

No 2D distribution of thermals

6

Our question: how are thermals arranged in a horizontal plane? The theory sorts clouds by type, not by position.

Spatial quantities (schematic)



Sought, not in the paper

R_i radius of thermal i at a fixed height

(x_i, y_i) position of its centre

d_{ij} distance between thermals i and j

Why the theory does not answer it

A spectrum over λ has no positions or distances.

$\mathcal{M}_B$ mixes cloud number, lifetime and single-cloud mass flux, never separated.

The environment is assumed horizontally uniform.

A mass-flux spectrum leaves the horizontal thermal field unspecified

Scope: §4 and §7. The sketch defines the quantities sought. It represents no observed or simulated thermal field.

Paper review: Arakawa & Schubert (1974) | Slide 6

No 2D distribution of thermals

The target of our reading was the distribution of thermals in a horizontal plane. The sketch defines sizes, positions and centre-to-centre separations. It is illustrative only. Arakawa and Schubert do not report distributions of these thermal quantities.

The model resolves cloud types through their fractional entrainment and vertical structure, while closing their collective mass-flux contribution. A spectrum over λ is not a two-dimensional spatial distribution. The paper explicitly separates mass flux from cloud number, lifetime and single-cloud transport, which are not determined independently (Eq. 145). Its use of a relation between entrainment and cloud radius does not yield a measured thermal-radius distribution.

Horizontal uniformity of the environment is a modelling assumption, not evidence for randomly or uniformly positioned thermal centres. The cloud sketch in Figure 1 likewise supplies no spatial statistics. Source: Sections 4 and 7, particularly the discussion of Equations (94) and (145).